Lecture 10 Solving Games Without Saddle Points

Lecture Plan:

1. Concept of a game without a saddle point
2. Solution of games
3. Illustrative examples

§1 Concept of a game without a saddle point

Recap: Zero-Sum Matrix Games and Saddle Points In a two-player zero-sum game, the gain of one player equals the loss of the other. Payoffs are typically represented by a matrix for Player A (the maximizer), while Player B (the minimizer) seeks to minimize A's payoff. If the maximin equals the minimax, the game has a saddle point and admits a pure strategy equilibrium.

Concept of a Game Without a Saddle Point A game is said to have **no saddle point** when the maximum of the row minima (maximin) is not equal to the minimum of the column maxima (minimax). In such cases, no pure strategy pair is stable: at least one player can improve their payoff by deviating. Consequently, equilibrium must be sought in **mixed strategies**, where players randomize over available actions.

Key implications:

- Pure strategies are insufficient for optimal play.
- The equilibrium is probabilistic.
- The value of the game lies strictly between the maximin and minimax.

Mixed Strategies: Definition and Rationale A **mixed strategy** is a probability distribution over a player's pure strategies. If Player A has strategies A1, A2, …, Am, a mixed strategy assigns probabilities p1, p2, …, pm such that $p_i \geq 0$ and $\Sigma p_i = 1$.

Rationale:

- Prevents the opponent from exploiting predictable behavior.
- Ensures equilibrium existence in finite zero-sum games (by the Minimax Theorem).

4. Expected Payoff and the Indifference Principle When players use mixed strategies, outcomes are evaluated via **expected payoff**. Let A mix with probabilities (p, 1−p) over A1 and A2, and B mix with (q, 1−q) over B1 and B2. The expected payoff is computed as the weighted average of matrix entries.

- **Indifference Principle**: In equilibrium, each player's mixed strategy makes the opponent indifferent among the pure strategies used with positive probability. This condition yields equations to solve for equilibrium probabilities.

§2 Solution of games

Analytical Solution of 2×2 Games

Consider the payoff matrix:

	B1	**B2**
A1	a	b
A2	c	d

If no saddle point exists, let Player A play A1 with probability p and A2 with (1−p). To make Player B indifferent:

Expected payoff if B plays B1: E(B1) = ap + c(1−p) Expected payoff if B plays B2: E(B2) = bp + d(1−p)

Set E(B1) = E(B2) and solve for p.

Similarly, solve for q (probability that B plays B1) by making A indifferent.

The **value of the game** V is obtained by substituting equilibrium probabilities into the expected payoff expression.

Graphical Method (2×n or m×2 Games)

When one player has two strategies and the other has more, a graphical approach can be used:

- Plot expected payoff lines as functions of the mixing probability.
- Identify the lower envelope (for the minimizing player) or upper envelope (for the maximizing player).
- The intersection point gives the optimal mixed strategy and the game value.

Linear Programming Formulation (m×n Games) General matrix games can be solved using linear programming.

For Player A: Maximize V subject to:

- Σ pi aij $\geq$ V for all j
- Σ pi = 1, pi $\geq$ 0

For Player B (dual problem): Minimize V subject to:

- Σ qj aij $\leq$ V for all i
- Σ qj = 1, qj $\geq$ 0

This formulation ensures optimal mixed strategies and the value of the game.

Illustrative Examples

Example 1 (2×2 game without saddle point):

	B1	B2
A1	2	5
A2	4	1

Row minima: 2, 1 → maximin = 2 Column maxima: 4, 5 → minimax = 4

No saddle point exists.

Solve using indifference: E(B1) = 2p + 4(1−p) E(B2) = 5p + 1(1−p)

Set equal: 2p + 4 − 4p = 5p + 1 − p 4 − 2p = 4p + 1 3 = 6p → p = 0.5

Similarly, q = 0.5. The value of the game V = 3.

Example 2 (Matching Pennies-type game):

	B1	B2
A1	1	-1
A2	-1	1

No saddle point exists. The equilibrium is p = q = 0.5, with value V = 0.

Games without saddle points require mixed strategy solutions. By introducing probabilistic behavior, players can achieve equilibrium even in adversarial environments. Analytical, graphical, and optimization-based methods provide systematic tools for solving such games.

Self-Monitoring Questions

1. What distinguishes a game without a saddle point from one with a saddle point?
2. Why are mixed strategies necessary in such games?
3. What is the indifference principle and how is it used?
4. How do you solve a 2×2 game analytically?
5. When is the graphical method applicable?
6. How can linear programming be used to solve matrix games?

Recommended Literature

1. Von Neumann, J., & Morgenstern, O. *Theory of Games and Economic Behavior*.
2. Osborne, M. J. *An Introduction to Game Theory*.
3. Myerson, R. B. *Game Theory: Analysis of Conflict*.
4. Gibbons, R. *A Primer in Game Theory*.
5. Rasmusen, E. *Games and Information*.
6. Binmore, K. *Playing for Real: A Text on Game Theory*.